

MASTERCLASS BAR CHARTS

The Journey from Data to Visual Intelligence

1. What is a Dataset?

A **dataset** is a collection of data arranged in rows and columns.

Example

Name	Sport	Study Hours
Aman	Cricket	2
Riya	Football	3
Anjali	Basketball	4

Diagram

Dataset

Name	Sport	Study Hours
Aman	Cricket	2
Riya	Football	3

2. Rows and Columns

Row

One complete record.

Aman | Cricket | 2

Column

One type of information.

Name
Aman
Riya
Anjali

Quick Rule

Row = One Record
Column = One Attribute

3. Types of Data

Categorical Data

Names or labels.

Examples

- Cricket
- Football
- India
- Red

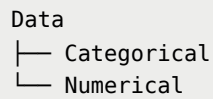
Numerical Data

Numbers that can be measured or counted.

Examples

- Height
- Weight
- Marks
- Age

Diagram



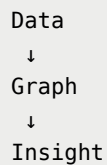
4. Why Do We Need Graphs?

Large amounts of data are difficult to understand.

Graphs help us:

- Compare values
- Find patterns
- Understand information quickly

Diagram



5. Data Visualization

Data Visualization is the process of representing data using graphs and charts.

Benefits

- Easy to understand
- Easy to compare
- Easy to communicate

Bar Graphs

6. What is a Bar Graph?

A bar graph is used to compare different categories.

Example

Sport	Students
Cricket	30
Football	20
Basketball	10

Diagram



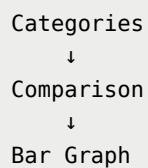
7. When to Use a Bar Graph?

Use a bar graph when comparing categories.

Examples

- Favourite sport
- Favourite subject
- Mobile brands
- Countries

Diagram



8. Rules of a Bar Graph

Rule 1

Data should be categorical.

✓ Sports

✓ Subjects

✗ Height

✗ Weight

Rule 2

Bars should have equal width.

Rule 3

Bars should have gaps between them.

Rule 4

Axes must be labelled.

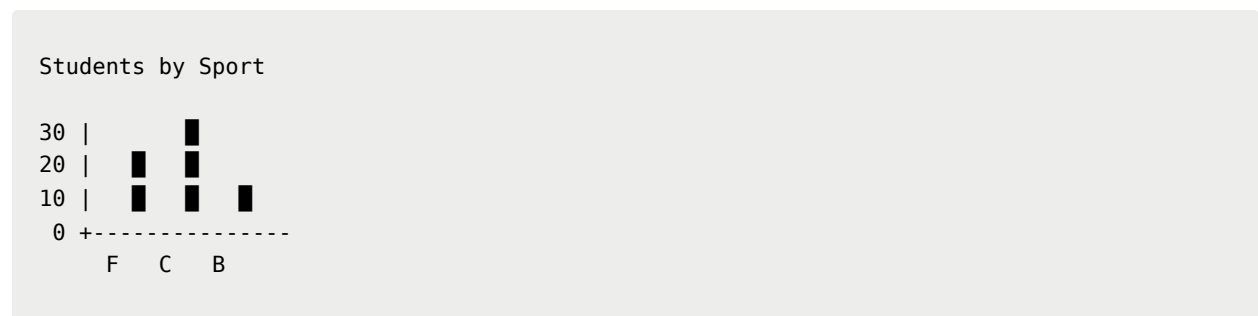
Rule 5

Graph should have a title.

Rule 6

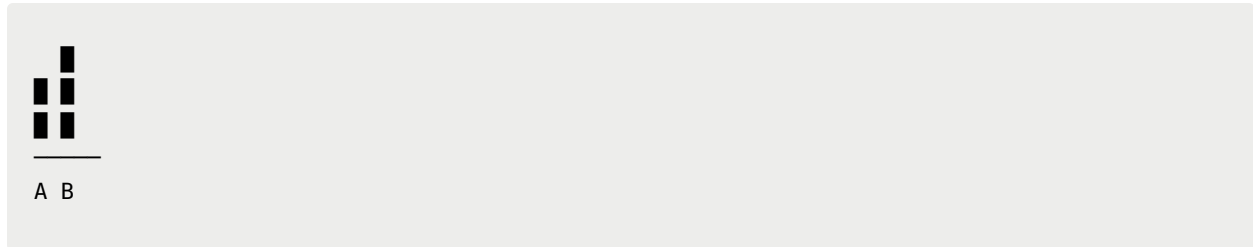
Y-axis should usually start from zero.

Diagram



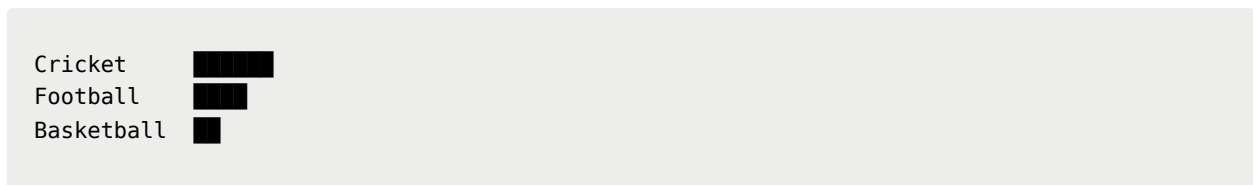
9. Types of Bar Graphs

Vertical Bar Graph



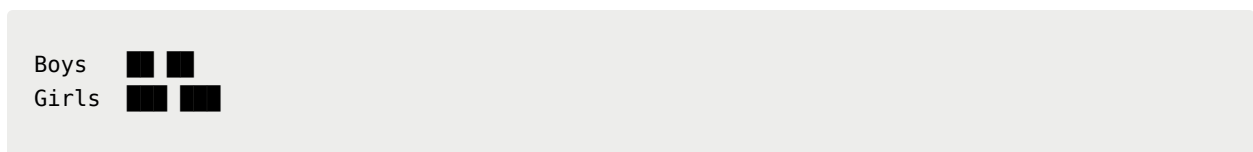
Used for simple comparisons.

Horizontal Bar Graph



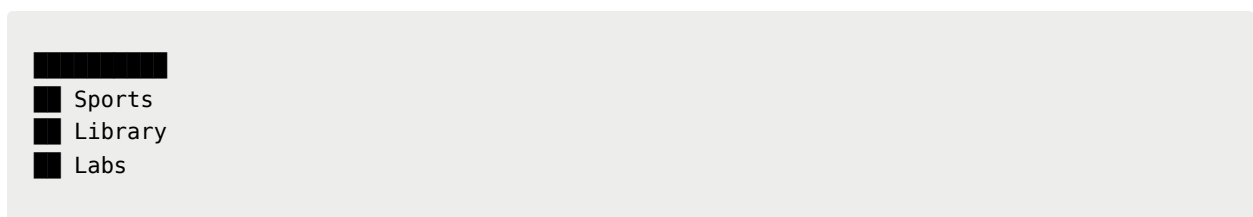
Used when labels are long.

Grouped Bar Graph



Used to compare multiple groups.

Stacked Bar Graph



Used to show parts of a total.

Histograms

10. What is a Histogram?

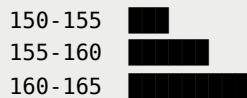
A histogram is used to visualize continuous numerical data.

Instead of individual values, data is grouped into ranges.

Example

Height Range	Students
150-155	4
155-160	8
160-165	12

Diagram



11. Why Do Histogram Bars Touch?

The ranges are continuous.

150 - 155
155 - 160
160 - 165

There are no gaps between ranges.

Diagram



12. Bar Graph vs Histogram

Bar Graph	Histogram
Categorical Data	Numerical Data
Bars Have Gaps	Bars Touch
Compare Categories	Show Distribution

Diagram

Bar Graph



Histogram



Summary

Dataset



Rows & Columns



Types of Data



Data Visualization



Bar Graphs



Histograms

Key Points

- Dataset = Collection of data
- Row = One record
- Column = One attribute

- Bar Graph = Compare categories
- Histogram = Analyze numerical ranges
- Visualization makes data easier to understand